BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 241 · Thinking procedurally: ordering subprocedures

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Why does `age>12 && age<18` reject two valid ages for a range 12–18 inclusive?

Q2. What is a subprocedure?

Q3. What is an object's state?

Q4. Why not send a success message before saving a transaction?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

decomposition — Split a problem into smaller operations.

subprocedure — Named step used within a larger solution.

dependency — Stage that must precede another.

state update — Change the stored record or value.

transaction — Coordinated operations that must succeed reliably.

New learning

Decompose the task — Separate the problem into input validation, checking availability, updating the record and notifying the user.

Identify dependencies — No booking should be saved until seats are available; no confirmation should be sent before a successful save.

Make smaller operations reusable — ValidateRequest and SendConfirmation can become independently testable subprocedures.

Worked example

Put password-reset stages in a safe order.

1. Predict first — First identify the account and create a single-use reset token.

2. Model the reasoning — Check the token before accepting a new password.

3. Model the reasoning — Save the change successfully before confirming to the user.

We Do · Practise with guidance

Order steps for borrowing a library book online.

```
Find book → Check available
→ Record loan → Update availability
→ Confirm
```

Predict first

Identify the book and verify availability before trying to record a loan.

Model the reasoning

Store the loan and change availability, then confirm only if those actions succeed.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Order these steps for ticketing: record purchase, take request, check payment, issue ticket.

Question 2

Design four named subprocedures for booking a classroom and give two key ordering requirements.

Question 3

A booking code subtracts seats from a local int parameter but never updates the central system. Identify three problems and three remedies.

Exit Ticket · Answer before checking

A booking confirmation can safely be sent before the record is saved.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Because strict comparisons exclude both 12 and 18.

Q2. A named, reusable piece of a larger task.

Q3. The stored values of its properties or fields.

Q4. The save might fail even though the user has already been told it succeeded.

You Do pages · Independent questions

Q1. [4] Take request [1], check payment [1], record purchase [1], issue ticket [1].

Q2. [6] ValidateInput [1], CheckRoomAvailable [1], SaveBooking [1], SendConfirmation [1]; check room before save [1], save before confirm [1].

Q3. [6] Parameter is a copy [1], central count unchanged [1], another user could book same seats [1]; update stored capacity [1], use an atomic save/check [1], confirm only after successful save [1].

Exit Ticket · Model answer

False. The save might fail, leaving an unconfirmed booking in reality despite the message.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first.

GREEN PEN Then: tick and rewrite mistakes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding